Antimatter and the Metaphysics of Quantity

Hans Halvorson
(with David Baker)
Princeton University

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Outline

Philosophical accounts of quantity

A brief history of antimatter

What antimatter could not be

Antimatter and the structure of quantities

Quantities and determinables

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Determinable	Determinate
color	red, green, blue, ...
mass	non-negative real number
position	point in space
isospin	$0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$

Proposal

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A quantity is a semi-group; i.e. a set S with a binary function $\star : S \times S \to S$.

Proposal (Denby)

A quantity is a pair $\langle S, f \rangle$ where S is a metric space, and f is a function from concrete particulars to S.

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- Denby's account is based on a pre-scientific notion of "similarity" between properties — e.g. "redness" is more similar to "purpleness" than it is to "blueness".

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- Ongoing: Apparent asymmetry of matter and antimatter in our universe is one of the greatest unsolved problems in contemporary physics.
- 1969–1990: Doplicher, Haag, and Roberts explicate the structure of fundamental physical quantities

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Other reason to care about antimatter: Asymmetry of matter/antimatter connected to asymmetry of future/past?

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- Not a necessary condition: Antiparticles do not always annihilate their antiparticles, but sometimes produce other particles
- Not a necessary condition: Some particles are their own antiparticles (e.g. photon, neutral pion)

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- Feynman's heuristic reasoning does not amount to an analysis.
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- The claim is not general enough: the antimatter concept is more general than the particle concept.

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Definition: A *wavefunction* ψ is an assignment of complex numbers to the determinates of a determinable.

A (nonzero) complex number is a pair consisting of a length, and an angle.

$$\psi(e) \quad = \quad (\text{probability}, \text{phase})$$

Complex Conjugation

$$(\text{probability}, \text{phase}) \quad \overset{\text{conjugate}}{\Longrightarrow} \quad (\text{probability}, -\text{phase})$$

Conjugate wavefunction

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- Wrong: the conjugate wavefunction is basis-dependent; cannot ground a fundamental physical relation.

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- Strictly false: some quantities' values are not reversed (e.g. position, velocity)
- Even false for "essential" quantities. (e.g. mass)
- The concept of "opposite value" is left unanalyzed
  On going philosophical accounts of quantity (e.g. Mundy) the specific assignment of real numbers to properties is conventional

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> *"...a quantum number ... is 'additive' if, in order to derive its value for a composite entity, we simply add up the individual values for the constituent particles"*

The opposite of a property

Newton's third law: For every force acting on an object, the object will exert an equal, yet *opposite*, force on its cause.

Opposites	No Opposites
	Temperature
Newtonian Forces	
Electric Charge	Position in post-Aristotelian space
Relative Position in STR	Relative Position in GTR
	Ounces of gold
Essential quantities in quantum field theory	

Abelian quantum numbers (e.g. electric charge)

- There are an infinite number of charges $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
- Charges can be **composed**, and the composition operation is modelled by addition of real numbers.
- There is a unique **neutral** charge, to which we assign value 0.

Nonabelian quantum numbers (e.g. isospin)

Isospin takes values in $\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$
Composition rules:

$$i \otimes 0 = 0 \otimes i = i$$

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

$$1 \otimes \frac{1}{2} = \frac{1}{2} \oplus 1 \oplus \frac{3}{2}$$

$$i \otimes j = \bigoplus_{k=|i-j|}^{i+j} k$$

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- Mathematical structure:
  $x = k_1 \oplus \cdots \oplus k_n$ means that $k_1, \ldots, k_n$ are "parts" of x, i.e. there are orthogonal embeddings of the objects k in the object x. (Each of the objects $0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$ is itself irreducible.)

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Conjugate objects

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- Electric charge: the conjugate of i is $-i$, and $i \neq -i$.
- Isospin: the conjugate of i is i.

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Examples:

- The symmetry group for isospin is $SU(2)$. In the dual of $SU(2)$, each object is self-conjugate; hence, each isospin quantum number is its own inverse.

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Problem: Why does this work? What does the dual of the symmetry group have to do with physical quantities?

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Ansatz (Doplicher, Haag, Roberts)

Each UIR corresponds to an essential determinate (value of a quantity) of elementary particles.

DHR program: derive structure of quantities

Question: Does the "set" of UIRs have an interesting additional structure — derivable from the physics?

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Answer: Yes. For any two representations π and π':

- There is a product representation $\pi \otimes \pi'$;
- There is a sum representation $\pi \oplus \pi'$;
- There is a unique identity representation π_0 such that $\pi \otimes \pi_0 = \pi = \pi_0 \otimes \pi$;
- There is a unique conjugate representation $\overline{\pi}$ such that $\pi \otimes \overline{\pi}$ and $\overline{\pi} \otimes \pi$ contain a single copy of π_0.

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- A *tensor category* is a category with a binary operation $\otimes$ on objects and arrows, and a tensor unit $1_{\mathcal{C}}$, that satisfy certain laws (e.g. associativity).
- A tensor category has *conjugates* just in case for each object X there is an object $\overline{X}$ such that $X \otimes \overline{X}$ and $\overline{X} \otimes X$ contain a single copy of the tensor unit $1_{\mathcal{C}}$.

De-mystifying the magic symmetry principle

Theorem (Doplicher-Roberts 1990)

The category of UIRs is equivalent to the dual of a symmetry group.

Metaphysical lessons

- Stop equating "quantity" with "something representable by real numbers" — that is just a *special case*. (Corollary: Rethink the contrast between "qualitative" and "quantitative".)

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- Contra Mundy and Denby: determinates do have structure, but look to fundamental physics to tell us what it is.
- Contra the trend of relativizing: "zero" is in some cases objective and absolute.